

Mathematics, 2018 EC, entrance exam, version 1

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 29

Total marks: 29

Name _____ Grade _____ ID No. _____ Date _____

1. After rationalizing the expression $\frac{3}{(3\sqrt{3} - 2\sqrt{2})}$, the denominator is equal to (1)
 - A. 19
 - B. 57
 - C. $\frac{3(3\sqrt{3} - 2\sqrt{2})}{19}$
 - D. $\frac{3(3\sqrt{3} + 2\sqrt{2})}{19}$
2. The solution set of the system $\{3x - y = 5, 3y + 15 = 9x\}$ is (1)
 - A. $\{(0,-5), (1,-2), (2,-1), \dots\}$
 - B. $\mathbb{R}$
 - C. $\emptyset$
 - D. $\{(t, 3t-5): t \in \mathbb{R}\}$
3. The solution set of $|2x+1| = x-7$ is (1)
 - A. $\{-8, -2\}$
 - B. $\{-8, 2\}$
 - C. $\{8, 2\}$
 - D. $\emptyset$
4. Suppose N is the set of natural numbers and $A = \{1, 2, \{3\}\}$, then which of the following is true? (1)
 - A. $\{3\} \subseteq A$
 - B. $N \cup A = N$
 - C. $A \setminus \{3\} = A$
 - D. $A \subseteq N$
5. Let $A = \{0, 1, 2, 3, 4\}$ and R be a relation from A to A given by $R = \{(x, y): y = x+2, \text{ for } x, y \in A\}$. Which one is true about R ? (1)
 - A. $(2, 3)$ is an element of R
 - B. $(3, 5)$ is an element of R
 - C. The domain of R is $\{1, 2, 3, 4\}$
 - D. The range of R is $\{2, 3, 4\}$
6. The graph of $f(x) = x^2 + 6x + 7$ is obtained from the graph of $g(x) = x^2$ by shifting (1)
 - A. 6 units to the right and 9 units upward
 - B. 3 units to the right and 2 units downward
 - C. 3 units to the left and 2 units downward
 - D. 3 units to the left and 2 units upward
7. Which one of the following points belongs to the relation $R = \{(x, y): y \geq 3x+1\}$? (1)
 - A. $(2, 1)$
 - B. $(0, 1)$
 - C. $(1, 3)$
 - D. $(-2, -6)$
8. For $f: \mathbb{R} \rightarrow A$, $f(x) = -x^2 - 6x + 5$ to be onto, what must A be? (1)
 - A. $(-\infty, 14]$
 - B. $(-\infty, -22]$
 - C. $[-22, \infty)$
 - D. $(-\infty, 0]$
9. What must be added to $x^3 + 5x^2 - 3x - 8$ so that $x+2$ is a factor? (1)
 - A. -10
 - B. 10
 - C. -14
 - D. 14

10. Which one of the following is always true about a polynomial function of degree 3? (1)
- A. It has three zeros
B. Its range is $\mathbb{R}$
C. It may not cross the x-axis
D. It crosses the x-axis three times
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11. The simplified form of $\log_{\{4x\}}\{4x^{-1}\} + 4 \log_{\{4x\}}\{2x^2\} - 4 \log_{\{4x\}} x$ is: (1)
- A. 3
B. 4
C. $1/4$
D. $\sqrt{4}$
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12. What is the simplified form of $(2^{\sqrt{5}} \times 2^{\sqrt{20}}) / 8^{\sqrt{5}}$? (1)
- A. 1
B. 2
C. $2^{\sqrt{5}}$
D. $\sqrt{2}$
-
13. Which one of the following is true about the graph of $f(x)=1+(1/2)^x$? (1)
- A. It is increasing on $[0, \infty)$
B. The x-axis is its asymptote
C. It intersects the y-axis at $(0,1)$
D. for $x>0$, it lies between the lines $y=1$ and $y=2$
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14. Let $f(x)=\log_{\{1/9\}} x$, then which one is true about $f(x)$? (1)
- A. f is an increasing function
B. $y=0$ is asymptote of f
C. for each $x \in (0,1)$, $f(x)>0$
D. it has y-intercept
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15. If $\log_x 64 + \log_x (1/8) = 3/5$, then x is equal to: (1)
- A. 32
B. 16
C. 8
D. 4
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16. Which one of the following is a power function? (1)
- A. $f(x) = (2/3)^{\sqrt{x}}$
B. $f(x) = (2/3)^x$
C. $f(x) = 2x^4 - 1$
D. $f(x) =$
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17. Which one of the following statement define the greatest integer function given by $f(x)=|x|$? Wait, the text says 'f(x)=|x|' but that's absolute value, not greatest integer. Probably a misprint. The options describe floor function. It says: 'Which one of the following statement define the greatest integer function given by $f(x)=|x|$?' Actually greatest integer is floor. Options: A. It is the least integer greater than or equal to x . B. It is an integer greater than or equal to x . C. It is an integer less than or equal to x . D. It is the greatest integer less than or equal to x . So D is definition of floor function. Answer D. (1)
- A. least integer $\geq x$
B. integer $\geq x$
C. integer $\leq x$
D. greatest integer $\leq x$
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18. The inverse of the function defined by $f(x) = 3x/(x+2)$ is (1)
- A. $f^{-1}(x) = -2x/(x-3)$
B. $f^{-1}(x) = 2x/(x-3)$
C. $f^{-1}(x) = -2x/(x+3)$
D. $f^{-1}(x) = 2x/(x+3)$
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19. If 20% salt solution is to be mixed with 40% salt to form a mixture of 30% salt solution of 600 liters, then how much of the 40% salt solution is needed? (1)
- A. 100 liters
B. 200 liters
C. 300 liters
D. 400 liters

20. For any $n \times n$ matrix A , which one is true? (1)
- A. If B is obtained by interchanging two rows of A , then $\det(A) = \det(B)$
B. If A is invertible, then $\det(A) = 1/\det(A^{-1})$
C. $\det(A) = -\det(A^t)$
D. If k is scalar, $\det(kA) = k \det(A)$
21. For what value of x is the matrix $A = \begin{bmatrix} x & -3 & x \\ -1 & 0 & 2 \\ 2 & 1 & -5 \end{bmatrix}$ non-invertible (singular)? (1)
- A. 1
B. -1
C. 2
D. -2
22. The 3×3 system $AX = B$ has solution $(2, b, -4)$. If $\det(A) = -2$ and the determinant of the matrix obtained by replacing the second column of A by B is $1/4$, then what is b ? (1)
- A. $-1/8$
B. -8
C. $-1/2$
D. -2
23. Let $\{a_n\}$ be arithmetic with $a_3 = 7$ and common difference -4 , then the n th term (general term) is: (1)
- A. $13 - 2n$
B. $16 - 3n$
C. $19 - 4n$
D. $22 - 5n$
24. A certain meeting hall has rows with 27, 31, 35, 39, ... seats, last row 107 seats. Total number of seats? (1)
- A. 1407
B. 1427
C. 1307
D. 1327
25. What is the local minimum value of $f(x) = x^3 - 12x + 1$? (1)
- A. -24
B. -20
C. -15
D. -14
26. Let $f(x) = 6x/(a-x)$. For what value of a is $f'(-2a) = 1/4$? (1)
- A. $3/8$
B. $8/3$
C. $5/4$
D. $4/5$
27. The time rate of change of the radius of a sphere is $1/(2\pi)$ cm/sec when radius is 5 cm. What is the rate of change of surface area? (1)
- A. $10 \text{ cm}^2/\text{sec}$
B. $20 \text{ cm}^2/\text{sec}$
C. $30 \text{ cm}^2/\text{sec}$
D. $40 \text{ cm}^2/\text{sec}$
28. What is the area of the region bounded by $f(x) = x^2 - 3x + 2$ and the x -axis on $[0, 2]$? (1)
- A. 1
B. $3/4$
C. $3/2$
D. $-3/4$
29. The interquartile range of the data 7, 3, 5, 1, 6, 1, 6, 8 is (1)
- A. 4.5
B. 4
C. 3.5
D. 3

— END OF EXAMINATION —